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AI / ML • ASSET MANAGEMENT • RISK INTELLIGENCE

AI-Powered Portfolio Risk Management at BlackRock

Using AI/ML and Aladdin to Enhance Investment Risk Analysis and Decision Support

BBA FinTech | Chitkara Business School

Course / Section: [TO BE ADDED] • Faculty / Instructor: [TO BE ADDED] • Date: [TO BE ADDED]

GROUP MEMBERS

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Note: 11 names were supplied for a group described as 10 members — please confirm the roster and remove the extra entry before final submission.

Company & Industry Overview

BlackRock, Inc.

- World's largest asset manager, investing on behalf of institutional and individual clients
- Manages multi-asset portfolios across equities, fixed income, and alternatives
- Serves pension funds, governments, corporations, financial advisors and individual investors
- Technology and data are core to its investment and risk-management process

The Asset-Management Industry

- Portfolios are increasingly complex, multi-asset and globally diversified
- Continuous risk management and market-data analysis are required
- Portfolio complexity is rising faster than manual analysis can scale
- Analytics and technology platforms are now central to competitiveness

Aladdin — BlackRock's Investment & Risk Platform

- Publicly documented role across investment management, portfolio analytics and risk management
- Supports data integration, scenario analysis and stress testing across asset classes
- Only documented, publicly-confirmed AI-enabled capabilities are referenced in this project



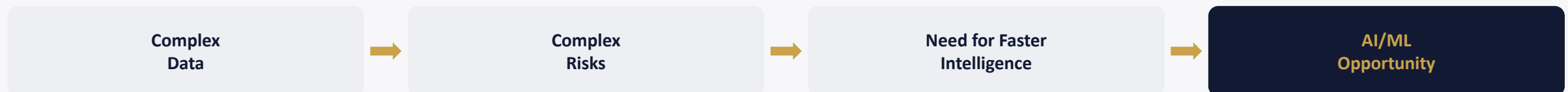
Business Problem

How can BlackRock identify and prioritize emerging portfolio risks faster when investment portfolios generate large volumes of complex, multi-asset and market data?

Why This Matters

- Large, complex investment portfolios across many asset classes
- High volume of fast-moving market data
- Rapidly changing market conditions
- Interconnected, hard-to-trace risk factors
- Need for continuous, real-time monitoring
- Need for robust scenario and stress analysis
- Difficulty manually examining every risk relationship
- Opportunity for AI/ML to support (not replace) analysts

Key message: BlackRock already has sophisticated risk analytics through Aladdin. The academic opportunity explored here is how additional AI/ML capabilities could enhance risk identification, prioritization and explanation.



Data Required

1. Portfolio Data

- Asset holdings & weights
- Security positions
- Asset allocation & exposures
- Cash positions

2. Market Data

- Equity prices, bond yields
- Interest & currency rates
- Commodity prices
- Volatility & credit spreads

3. Risk Data

- Historical volatility & correlations
- Risk factors & exposures
- Stress-test & scenario results
- Historical drawdowns

4. Macroeconomic Data

- Inflation & GDP indicators
- Interest-rate decisions
- Employment indicators
- Economic growth data

5. Unstructured Data

- Company filings
- Economic reports
- Central-bank communications
- Relevant news & research

6. Historical Data

- Historical market events
- Previous portfolio behavior
- Previous stress scenarios
- Historical model outputs

Data Requirements



Sensitive client and portfolio data requires strong access controls and governance throughout.

AI/ML Solution

AI-Enhanced Portfolio Risk Intelligence — Components

1

Machine Learning

Anomaly detection, risk-pattern identification, risk classification, predictive analytics, portfolio behavior analysis

2

NLP / Generative AI

Natural-language portfolio queries, risk explanations, market-event and scenario summaries

3

Existing Analytics Layer

Builds on existing portfolio and risk analytics where publicly documented

4

Human Expertise

Portfolio managers and risk professionals review and interpret every AI-generated insight

Proposed Architecture

1 Portfolio + Market Data

2 Data Processing

3 ML Risk Detection

4 Scenario / Stress Analysis

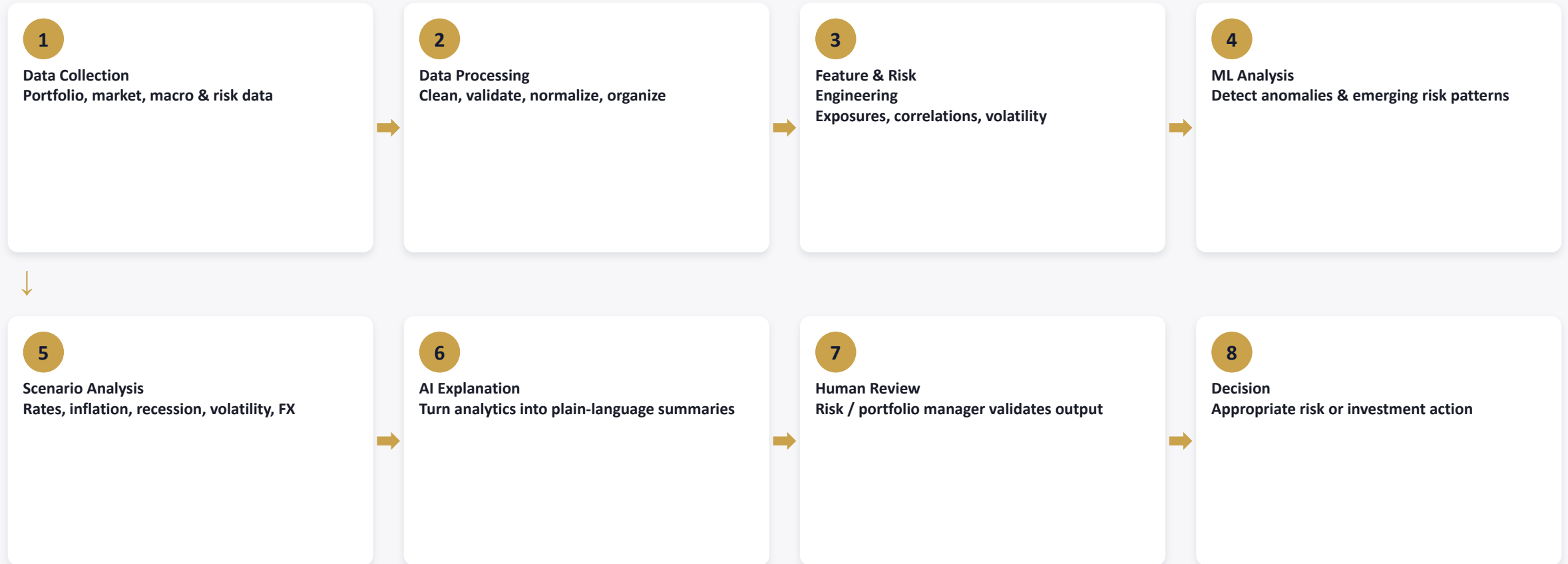
5 AI Explanation Layer

6 Risk Dashboard / NL Interface

7 Human Review

8 Investment / Risk Decision

How the AI Solution Works



AI SUPPORTS THE DECISION — HUMANS REMAIN ACCOUNTABLE

Business Impact

EXPECTED / PROPOSED BENEFIT

Operational Benefits

- Faster analysis & automated monitoring
- Reduced manual effort
- Faster access to portfolio insights
- Greater scalability

Risk Management Benefits

- Earlier identification of unusual patterns
- Better portfolio visibility
- More systematic scenario analysis
- Improved risk prioritization

Decision-Support Benefits

- Natural-language access to analytics
- Easier interpretation of risk information
- Faster analyst workflows
- Better communication of risk

Client / Business Benefits

- More timely reporting
- Better transparency
- More responsive risk analysis
- Stronger client confidence

AI/ML Capability



Business Effect



Expected Benefit

No numerical percentages are asserted — all figures shown are illustrative labels, not claimed metrics.

Risks, Ethics & Responsible AI

Risk Area	Risk	Mitigation
Data Privacy	Sensitive client & portfolio information	Access controls, encryption, permissioning, audit trails
Model Bias	Historical data may embed bias or limitations	Bias testing, model monitoring, dataset evaluation
AI Hallucination	Generative AI may produce inaccurate output	Grounded/retrieval-based systems, output validation
Explainability	Users may not understand AI risk flags	Explainable AI, feature importance, auditability
Overreliance on AI	Users may accept AI output uncritically	Human-in-the-loop, approvals, training
Cybersecurity	AI systems may be targeted or manipulated	Security monitoring, access controls, governance

PRIVACY

FAIRNESS

TRANSPARENCY

ACCOUNTABILITY

HUMAN OVERSIGHT

AI Tools & Prompts Used

How We Used AI in This Project

1

Research

AI-assisted research and organization of source material

2

Prompt Engineering

Developing and refining structured prompts

3

Content Generation

Drafting explanations and slide content

4

Fact Verification

Cross-checking AI claims against reliable sources

5

Presentation Design

Tools actually used for slide design and layout

Example Prompts Used

Research Prompt

"Analyze BlackRock's publicly documented Aladdin capabilities relevant to portfolio risk management, identify the business problem, relevant data and AI/ML opportunities, and clearly separate documented facts from proposed solutions."

AI/ML Prompt

"Design a proposed AI/ML solution to enhance portfolio risk monitoring, focused on anomaly detection, scenario analysis and decision support, clearly flagging proposed vs. documented components."

Responsible AI Prompt

"Identify the major privacy, bias, explainability, hallucination, cybersecurity and human-oversight risks of applying AI to portfolio risk management in asset management."

Prompt Improvement & AI Evaluation

BEFORE

“Tell me about BlackRock and how it uses AI.”

- Too broad, no defined role
- No business problem or context
- No source or output requirements
- Facts and assumptions could mix

AFTER

“Act as a financial-services AI/ML consultant. Analyze BlackRock's documented use of technology in risk management, identify data, proposed enhancements, impact and responsible-AI risks — clearly separating fact from proposal.”

- Expert role + full context
- Explicit research & source constraints
- Defined output structure

What Changed

Basic Prompt	Improved Prompt
Broad	Specific
No role	Expert role
Little context	BlackRock + Aladdin context
No constraints	Research constraints
No verification	Source requirement
No structure	Defined output

Learning: Adding role, context, constraints, source requirements and output structure produced more focused, better-verified research — though AI-generated information still required human verification before inclusion.

Recommendations & Conclusion

Recommendations

- 1 Use AI as a decision-support tool, not an autonomous investment decision-maker
- 2 Maintain human-in-the-loop review for all material risk and investment decisions
- 3 Implement strict data governance and permission-based access
- 4 Continuously monitor models for accuracy, bias, security and drift
- 5 Provide explainable AI outputs so professionals can question model-generated insights

Conclusion

BlackRock operates in a data-intensive investment-management environment where portfolio complexity makes timely risk analysis essential. Its Aladdin ecosystem provides a strong foundation for investment and risk analytics. A proposed AI/ML enhancement could add anomaly detection, risk prioritization, natural-language interaction and automated insight generation — while maintaining human oversight and responsible-AI controls.

“The goal is not to replace investment professionals with AI, but to help them analyze complex information faster, understand risk more clearly, and make better-informed decisions.”

Learning Reflection, References & GitHub

1. LEARNING REFLECTION

- Learned how AI/ML connects to a real financial-services business problem
- Learned that data quality, governance and human oversight are critical in financial AI applications
- Learned that prompt engineering significantly improves the structure and relevance of AI-assisted research

AI/ML Learning:

AI/ML can support complex financial analysis by identifying patterns and processing large datasets, but appropriate controls and human oversight remain essential.

Prompt Engineering Learning:

Adding role, context, constraints, source requirements and output structure produces more focused AI responses.

Verification Example:

[Add one real example: an AI-generated claim the team checked against an official source, and how it was confirmed or corrected.]

Do not invent this example — replace with your team's actual research check.

2. REFERENCES

- BlackRock — official corporate website
- BlackRock Aladdin — platform overview
- BlackRock Aladdin Risk
- BlackRock Aladdin Copilot
- BlackRock annual report / investor relations
- Relevant academic & industry sources on AI in asset management

Full reference list with links available on request / via QR code.



3. GITHUB REPOSITORIES

Chanpreet	github.com/chanpreetsingh097
Akshit	github.com/akshit2522bbafai25-stack
Devansh	github.com/Devansh5203
Gursehaj	github.com/gursehajalag-cmyk
Abhay Sharma	github.com/abhay2505bbafai25-cpu
Athrav Chauhan	github.com/athravchauhan33-blip
Dushant Wadhwa	github.com/dushantwadhwa084-beep
Emon Sethi	github.com/sudo2233-hub
Harshit Saini	github.com/harshits7118-dot
Husanbir Singh Ahluwalia	github.com/Husanbir13
Akshay	github.com/aroraakshay190-del

Note: 11 repository links supplied for a stated 10-member group, and the source data flagged an extra/ambiguous URL near Devansh / Gursehaj. All entries are preserved as given — confirm roster & mapping before final submission.